RIGHT-HALF-PLANE ZERO REMOVAL TECHNIQUE FOR LOW-VOLTAGE LOW-POWER NESTED MILLER COMPENSATION CMOS AMPLIFIER

Ka Nang Leung, Philip K.T. Mok and Wing-Hung Ki

Department of Electrical and Electronic Engineering
The Hong Kong University of Science and Technology
Clear Water Bay, Hong Kong, China
Tel:(852)2358-8517 Fax:(852)2358-1485 E-mail:eemok@ee.ust.hk

ABSTRACT

The stability of a low-power CMOS three-stage nested Miller compensated (NMC) amplifier is deteriorated by a right-half-plane (RHP) zero. A technique using only one null resistor in the NMC amplifier to eliminate the RHP zero is developed. Both theory and experimental result show that the RHP zero is effectively eliminated by the proposed technique. The NMC amplifier using this technique has about 30% improvement on the bandwidth, slew rate and settling time with an improvement on the phase margin and no increase on the power consumption. Moreover, there is at least 49dB enhancement on the negative power supply rejection ratio.

1. INTRODUCTION

The trend of low-voltage design is an unjustified fact in today's circuit design world, and the demand of low-voltage cascade amplifiers in analog systems is increasing. Therefore, frequency compensation technique to stabilize multiple-stage amplifiers becomes more important. Nested Miller compensation (NMC) is a well-known frequency compensation technique, and it is good for stabilizing general-purpose operational amplifiers.

Although NMC can stabilize multiple-stage amplifiers, the NMC amplifiers considered in the literature are having high output current capability [1], [2] and the amplifiers are not supposed for low-power CMOS design. As will be shown in the paper, in low-power CMOS design, the stability of the NMC amplifier is deteriorated by a right-half-plane (RHP) zero [3], [4]. As a result, the closed loop stability of a low-power CMOS NMC amplifier is degraded.

In order to enhance the stability of an NMC amplifier, a technique called NMCNR, which is similar to the case of a two-stage amplifier, using a null resistor on the structure of the NMC amplifier to eliminate the RHP zero is presented in this paper. Moreover, as an amplifier with three gain stages can generally provide sufficient DC voltage gain (~100dB) with a good compromise on the power consumption, the discussion in this paper is focused on three-stage NMC amplifiers.

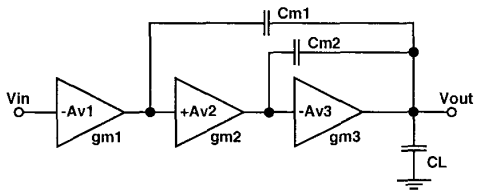


Fig. 1: Structure of the three-stage NMC amplifier

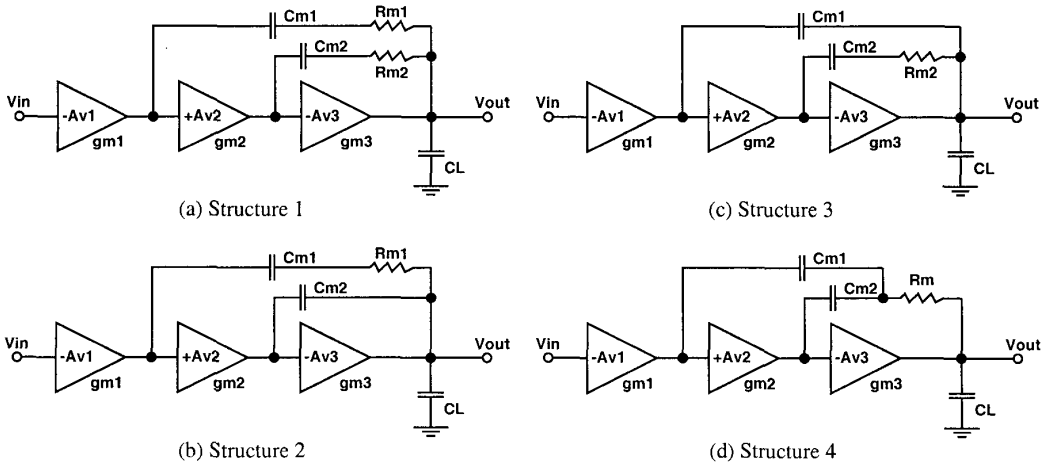
The organization of this paper is as follows. In the next section, the stability problems of nested Miller compensation in low-power CMOS design are presented. In section 3, four different possible structures of NMC amplifier using null resistor to eliminate the RHP zero are discussed, and an optimum structure is obtained from the analysis. The proposed structure NMCNR is presented in section 4 with detail theoretical analysis. Experimental results which show the improvement on both stability and performances of the NMCNR amplifier are included in section 5. Finally, conclusion is given.

2. STABILITY PROBLEMS OF NMC IN LOW-POWER CMOS DESIGN

In this section, the stability problems of NMC in low-power CMOS design will be illustrated by a three-stage NMC amplifier. The structure of a three-stage NMC amplifier is shown in Fig. 1. The voltage gains of the gain stages are $A_{v(1-3)}$. The capacitors C_{m1} and C_{m2} are the compensation capacitors, and C_L is the loading capacitor. For the structure of the NMC amplifier, the small-signal voltage gain transfer function, $A_v(s)=V_{out}(s)/V_{in}(s)$, is given by

$$A_v(s) = \frac{A_{DC}\left(1 - s\frac{C_{m2}}{g_{m3}} - s^2\frac{C_{m1}C_{m2}}{g_{m2}g_{m3}}\right)}{\left(1 + \frac{s}{p_{-3dB}}\right)\left[1 + s\frac{C_{m2}(g_{m3} - g_{m2})}{g_{m2}g_{m3}} + s^2\frac{C_L C_{m2}}{g_{m2}g_{m3}}\right]} \quad (1)$$

0-7803-5682-9/99/$10.00©1999 IEEE.



(a) Structure 1 (c) Structure 3

(b) Structure 2 (d) Structure 4

Fig. 2: Four possible structures

where $A_{DC}=g_{m1}g_{m2}g_{m3}R_1R_2R_3$ is the DC gain of the amplifier, $p_{-3dB}=(C_{m1}g_{m2}g_{m3}R_1R_2R_3)^{-1}$ is the dominant pole, and $g_{m(1-3)}$ and $R_{(1-3)}$ are the transconductance and the output resistance of the gain stages, respectively. From (1), it shows that there are three poles and two zeros. If the amplifier has high quiescent current in the output stage, g_{m3} is large and the zeros, which are located at high frequencies, can be neglected. However, since g_{m3} is small and may be comparable to g_{m1} and g_{m2} in low-power CMOS design, the effect of the zeros cannot be ignored.

From the numerator of (1), since the s and s^2 terms are both negative, the RHP zero locates at a lower frequency than the left-half-plane (LHP) zero and the effect of the RHP zero dominates. As a result, the phase margin of the NMC amplifier is reduced, and the closed loop stability of the amplifier in low-power CMOS design is degraded.

3. FOUR POSSIBLE STRUCTURES USING NULL RESISTOR TO ELMINATE THE RHP ZERO

The RHP zero of the NMC amplifier is introduced by the direct feed-forward capacitive paths, C_{m1} and C_{m2}, from the outputs of the input and intermediate gain stages to the output of the amplifier. A null resistor connected in series with the compensation capacitor can increase the impedance of the direct paths at high frequency, and this can reduce the feed-forward current. As a result, the RHP zero can be moved to a higher frequency or completely eliminated with an appropriate value of the null resistor.

Four different possible structures to eliminate the RHP zero in a three-stage NMC amplifier are shown in Fig. 2(a) to (d), respectively. Let $Z_{(1-4)}(s)$ be the functions of the zeros of the possible structures 1 to 4, by solving the circuit networks without losing accuracy, $Z_{(1-4)}(s)$ are given by

$$Z_1(s) = 1+s\left(C_{m1}R_{m1}+C_{m2}R_{m2}-\frac{C_{m2}}{g_{m3}}\right)+s^2\frac{C_{m1}C_{m2}(g_{m2}g_{m3}R_{m1}R_{m2}-g_{m2}R_{m1}-1)}{g_{m2}g_{m3}}$$

$$Z_2(s) = 1+s\left(C_{m1}R_{m1}-\frac{C_{m2}}{g_{m3}}\right)-s^2\frac{C_{m1}C_{m2}(g_{m2}R_{m1}+1)}{g_{m2}g_{m3}}$$

$$Z_3(s) = 1+s\left(C_{m2}R_{m2}-\frac{C_{m2}}{g_{m3}}\right)-s^2\frac{C_{m1}C_{m2}}{g_{m2}g_{m3}}$$

$$Z_4(s) = 1+s\left[C_{m1}R_m+C_{m2}\left(R_m-\frac{1}{g_{m3}}\right)\right]+s^2\frac{C_{m1}C_{m2}(g_{m3}R_m-1)}{g_{m2}g_{m3}}$$

(2)-(5)

In order to ease the prediction of the position of the zeros and eliminate the RHP zero at the same time, we want to minimize the number of the zeros. For the second-order functions shown in (2) to (5), this can be achieved by ensuring the s term to be positive and the s^2 term to be zero. As a result, only one LHP zero which increases the phase margin of the NMC amplifier is left.

As (2) is complex, it is very complicated to find out optimum values of R_{m1} and R_{m2} to eliminate the RHP zero when using structure 1. Moreover, matching of R_{m1} and R_{m2} is required. Thus, it is not a good choice. From (3) and (4), the RHP zero cannot be eliminated as the s^2 term is always negative for any value of R_{m1} and R_{m2}. Therefore, structure 2 and 3 are also not suitable.

For the structure 4, the s^2 term can be adjusted to zero by setting $R_m=1/g_{m3}$. When R_m is equal to the inverse of g_{m3}, the amplifier has no RHP zero and it only has a LHP zero which helps to increase the phase margin.

From the above analysis, it is concluded that structure 4 is the best among the four structures to eliminate the RHP zero of the amplifier, and the structure 4 will be used for the discussion in the next section.

4. NESTED MILLER COMPENSATION USING NULL RESISTOR

From the previous section, we know that structure 4 is the optimum frequency compensation network to eliminate the RHP zero effect. In this section, the frequency compensation technique using this structure called NMCNR will be presented.

By solving the circuit network without losing accuracy, the small-signal transfer function of the NMCNR amplifier with $R_m=1/g_{m3}$ is calculated as

$$A_v(s) = \frac{A_{DC}\left(1+s\frac{C_{m1}}{g_{m3}}\right)}{\left(1+\frac{s}{p_{-3dB}}\right)\left[1+s\frac{C_{m2}(g_{m3}-g_{m2})}{g_{m2}g_{m3}}+s^2\frac{\left(1-\frac{g_{m2}}{g_{m3}}\right)C_LC_{m2}}{g_{m2}g_{m3}}\right]} \quad (6)$$

Defining $k_g=g_{m2}/g_{m3}$, the transfer function shown in (6) is re-written as

$$A_v(s) = \frac{A_{DC}\left(1+s\frac{C_{m1}}{g_{m3}}\right)}{\left(1+\frac{s}{p_{-3dB}}\right)\left[1+s\frac{(1-k_g)C_{m2}}{g_{m2}}+s^2\frac{(1-k_g)C_LC_{m2}}{g_{m2}g_{m3}}\right]} \quad (7)$$

The stability of the NMCNR amplifier is obtained by considering that the poles of the amplifier has third-order Butterworth frequency response [5] in unity-gain configuration, and this is achieved by the following two dimension conditions for C_{m1} and C_{m2}.

$$C_{m1} = 4\left(\frac{g_{m1}}{g_{m3}}\right)C_L \quad (8)$$

$$C_{m2} = \frac{2}{1-k_g}\left(\frac{g_{m2}}{g_{m3}}\right)C_L \quad (9)$$

Using (8) and (9), the gain-bandwidth product (GBW) and the phase margin (PM) are given by

$$GBW = \frac{g_{m1}}{C_{m1}} = \frac{1}{4}\left(\frac{g_{m3}}{C_L}\right) \quad (10)$$

$$PM = 60^o + \tan^{-1}\left(\frac{z_{LHP}}{GBW}\right) > 60^o \quad (11)$$

where $z_{LHP}=g_{m3}/C_{m1}$.

As shown from (10), the GBW is controlled by C_{m1}, and from (7), the positions of the second and third pole are controlled by C_{m2}. Since it is generally not requiring an amplifier with a phase margin greater than 60°, we can reduce C_{m1} to increase the GBW and reduce the die size at the same time. Moreover, the slew rate (SR) of the NMCNR amplifier is better since the compensation capacitors are smaller and the time required to charge or discharge the compensation capacitors is shorter. As the settling time (T_s) is composed of the slewing and quasi-linear period [6], the overall settling behaviour of the NMCNR amplifier is also improved.

5. EXPERIMENTAL RESULTS

Low-voltage low-power NMCNR amplifier shown in Fig. 3 and the NMC counterpart were fabricated using a double-metal double-poly 0.8μm CMOS process from AMS (Austria Miko Systeme Group, Austria) with optimum stability conditions. The first, second and third gain stage are implemented by M101-M109, M201-M204 and M301-M302, respectively. The micrograph of the amplifiers is shown in Fig.4. Both amplifiers have a supply voltage of +/-1V and a load of 100pF in parallel with 25kΩ. The frequency responses of the NMCNR amplifier were measured by HP4194A impedance/gain-phase analyzer, and the frequency response of the NMCNR amplifier is shown in Fig. 5. The transient responses were measured by LeCroy 9354A oscilloscope and are shown in Fig. 6. The performances of the both amplifiers is summarized in Table 1.

The DC gain of both amplifiers are greater than 100dB, and the power consumption are the same. For the AC response, comparing the NMCNR amplifier to the NMC amplifier, both the GBW and PM are improved by 38% and 3°, respectively. Thus, NMCNR provides a wider bandwidth and better stability. For the transient response, the slew rate and settling time were measured in unity-gain configuration with a 0.5V step input. As shown in Table 1 and Fig. 6, the slew rate is enhanced by about 34%, and the settling time is reduced by 29%. Thus, NMCNR can settle the output voltage with shorter time.

Both the positive and negative power supply rejection ratio (PSRR) have significant improvement by NMCNR. It is due to NMCNR having smaller compensation capacitors and larger high frequency input-to-output voltage gain. Moreover, the null resistor helps to block the supply noise at high frequency.

The die of the NMCNR amplifier is smaller than that of the NMC amplifier. It is due to the smaller values of the compensation capacitors used in NMCNR, and they can be integrated by poly-poly capacitors with smaller area.

6. CONCLUSION

A novel frequency compensation technique NMCNR which eliminates the RHP zero in an NMC amplifier has been presented and proved by experimental results. The technique can improve the frequency and transient responses of low-voltage low-power CMOS NMC amplifiers for analog systems without an increase on circuit complexity and power consumption. Moreover, the area occupied by the NMCNR amplifier is smaller than that of the NMC amplifier.

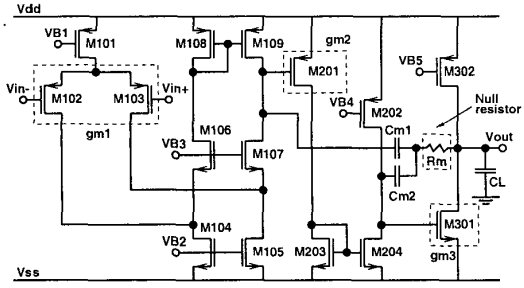


Fig. 3: Circuit diagram of the three-stage NMCNR amplifier

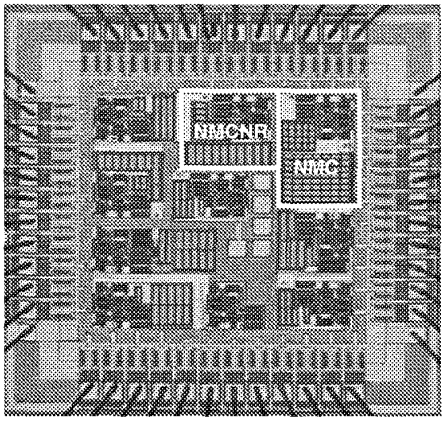


Fig. 4: Micrograph of the amplifiers

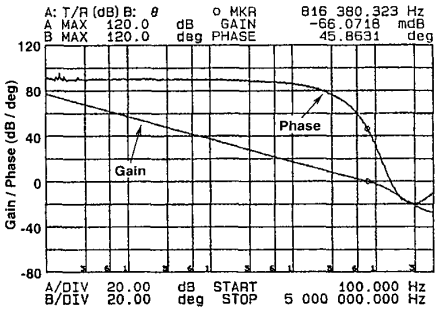


Fig. 5: Measured frequency response of the NMCNR amplifier

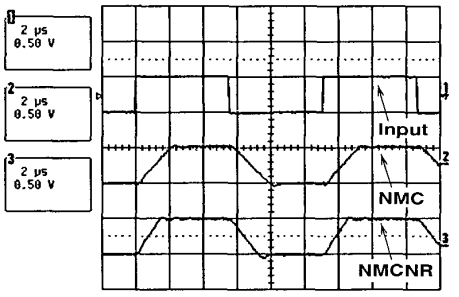


Fig. 6: Measured transient responses

Table 1: Measured results of the NMC and NMCNR amplifier with loading condition 100pF//25kΩ

	NMC	NMCNR
DC Gain	>100dB	
GBW	590kHz	816kHz
PM	43°	46°
SR^+/SR^- (V/µs)	0.23/0.23	0.36/0.31
T_s^+/T_s^- (µs)	4.25/4.36	3.03/3.04
$PSRR^+$@1kHz	85.80dB	92.61dB
$PSRR^+$@10kHz	64.10dB	75.62dB
$PSRR^-$@1kHz	53.66dB	106.44dB
$PSRR^-$@10kHz	35.61dB	84.62dB
$P_{consumption}$	400µW	
Power Supply	+/-1V	
Area	0.23mm^2	0.18mm^2

Note: Slew rate and settling time were measured at unity-feedback configuration with 0.5V step input

ACKNOWLEDGMENT

The work was supported by the RGC Competitive Earmarked Research Grant, Hong Kong SAR Government, HKUST6007/97E.

REFERENCE

[1] R.G.H. Eschauzier, L.P.T. Kerklaan and J.H. Huijsing, "A 100-MHz 100-dB Operational Amplifier with Multipath Nested Miller Compensation Structure", *IEEE Journal of Solid-State Circuits*, vol. 27, pp.1709-1717, 1992.

[2] S. Pernici, G. Nicollini and R. Castello, "A CMOS Low-Distortion Fully Differential Power Amplifier with Double Nested Miller Compensation", *IEEE Journal Solid-State Circuits*, vol. 28, pp.758-763, 1993.

[3] P.R. Gray and R.G. Meyer, *Analysis and Design of Analog Integrated Circuit*, 2nd edition. New York: Wiley, 1984.

[4] R. Gregorian and G.C. Temes, *Analog MOS Integrated Circuits for Signal Processing*, 1st edition. New York: Wiley, 1986.

[5] G.C. Temes and J.W. LaPatra, *Introduction to Circuit Synthesis and Design*, 1st edition. USA:McGraw-Hill, 1977.

[6] B.Y. Kamath, R.G. Meyer and P.R. Gray, "Relationship between Frequency Response and Settling Time of Operational Amplifier", *IEEE Journal of Solid-State Circuits*, vol. 9, pp.347-352, 1974.